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SEMANTIC PARSING OF UTTERANCE USING CONTRACTIVE PARAPHRASING

Abstract

Systems and methods are provided for automatically generating a program based on a natural language utterance using semantic parsing. The semantic parsing includes translating a natural language utterance into instructions in a logical form for execution. The methods use a pre-trained natural language model and generate a canonical utterance as an intermediate form before generating the logical form. The natural language model may be an auto-regressive natural language model with a transformer to paraphrase a sequence of words or tokens in the natural language utterance. The methods generate a prompt including exemplar input/output pairs as a few-shot learning technique for the natural language model to predict words or tokens. The methods further use constrained decoding to determine a canonical utterance, iteratively selecting sequence of words as predicted by the model against rules for canonical utterances. The methods generate a program based on the canonical utterance for execution in an application.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/229,637, filed on Apr. 13, 2021, the entire disclosure of which is hereby incorporated by reference.

BACKGROUND

[0002] Traditional systems recognize a natural language utterance for generating a command for execution. In practice, these traditional systems recognize a specific set of predefined natural language utterance that is in a particular task domain. In some of these systems, the recognition process relies upon a set of predefined mapping between the natural language utterance and one or more commands. Some other systems use natural language models to predict utterances based on given input. Developing a task-specific natural language processing systems requires generating extensive sets of rules and mapping data for training the natural language model. As areas of applications of computing systems expand, the needs of developing the natural language models for new areas of applications with new tasks arise. Thus, developing a technology that serves the needs and minimizes trade-offs in development costs would be desirable.

[0003] It is with respect to these and other general considerations that the aspects disclosed herein have been made. Also, although relatively specific problems may be discussed, it should be understood that the examples should not be limited to solving the specific problems identified in the background or elsewhere in this disclosure.

SUMMARY

[0004] According to the present disclosure, the above and other issues are resolved by semantic parsing of a natural language utterance based on a canonical utterance using a pre-trained, large-scale natural language model. The semantic parsing includes translating a natural language sentence (i.e., utterance) into a structured representation (e.g., program instructions for execution). In particular, the semantic parsing in the present disclosure receives the natural language utterance and generates a prompt. The prompt includes a set of the natural language utterance and exemplar conversational sentences that are relevant to the natural language utterance. A pre-trained natural language model predicts an utterance based on the prompt. In aspects, a “canonical utterance” includes a natural language-like sentence based on predefined grammatical rules and other rules for mapping to instructions. A constrained decoding generates a canonical utterance based on a sequence of words or tokens (i.e., parts of a word) as predicted by the natural language model. The predefined grammatical rules may enable efficient compilation of the canonical utterance into instructions (i.e., a logical form) for execution and/or for further processing.

[0005] The natural language model may be a pre-trained, auto-regressive natural language model

using a transformer. The natural language model receives the prompt. In aspects, the natural language model outputs a series of a set of words (and/or parts of a word as tokens) with probability values as predicted words based on billions of words of training data. A constrained decoder iteratively determines a sequence of words based on the output from the natural language model and generates a canonical utterance. The constrained decoder helps accurately generate that canonical utterance that can be compiled into instructions efficiently.

[0006] The instructions include one or more commands with parameters as specified by the natural language utterance through the canonical utterance. The automatic generation of the instructions based on the semantic parsing using the natural language model may be used in the area of a digital assistant system and a smart speaker, for example to provide various application services (e.g., a calendar application and a task management application). The process for generating a canonical utterance using contractive paraphrasing of a pre-trained natural language model with billions of trained parameters, followed by generating an instruction based on the canonical utterance will be the primary focus of the present disclosure.

[0007] This Summary is provided to introduce a selection of concepts in a simplified form, which is further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Additional aspects, features, and/or advantages of examples will be set forth in part in the following description and, in part, will be apparent from the description, or may be learned by practice of the disclosure.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0008] Non-limiting and non-exhaustive examples are described with reference to the following figures.

[0009] FIG. 1 illustrates an overview of an example system for performing semantic parsing for automatically generating an instruction in accordance to aspects of the present disclosure.

[0010] FIG. 2 illustrates an example of processing utterance based on semantic parsing in accordance with aspects of the present disclosure.

[0011] FIG. 3 illustrates an example of processing utterance based on semantic parsing in accordance with aspects of the present disclosure.

[0012] FIG. 4 illustrates an example of processing utterance based on semantic parsing in accordance with aspects of the present disclosure.

[0013] FIG. 5 illustrates an example of a method for semantic parsing in accordance with aspects of the present disclosure

[0014] FIG. 6 illustrates an example of a method for generating an instruction and performing the instruction based on semantic parsing in accordance with aspects of the present disclosure.

[0015] FIG. 7 is a block diagram illustrating example physical components of a computing device with which aspects of the disclosure may be practiced.

[0016] FIG. 8A is a simplified diagram of a mobile computing device with which aspects of the present disclosure may be practiced.

[0017] FIG. 8B is another simplified block diagram of a mobile computing device with which aspects of the present disclosure may be practiced.

DETAILED DESCRIPTION

[0018] Various aspects of the disclosure are described more fully below with reference to the accompanying drawings, which from a part hereof, and which show specific example aspects. However, different aspects of the disclosure may be implemented in many different ways and should not be construed as limited to the aspects set forth herein; rather, these aspects are provided

so that this disclosure will be thorough and complete, and will fully convey the scope of the aspects to those skilled in the art. Aspects may be practiced as methods, systems or devices. Accordingly, aspects may take the form of a hardware implementation, an entirely software implementation or an implementation combining software and hardware aspects. The following detailed description is, therefore, not to be taken in a limiting sense.

[0019] In examples, a semantic parser translates a natural language sentence into a structured representation or a logical form (e.g., a program, instructions, etc.). In further examples, semantic parsers may generate a canonical utterance as an intermediary before generating program instructions through various means, including using a predetermined set of grammatical rules for the canonical utterance. In aspects, the semantic parser may be a machine learnt semantic parser, which uses a trained machine learning model to generate the canonical utterance. The semantic parsers may score candidate canonical utterances based on relevance to the natural language utterance. The scored candidate canonical utterances may be filtered to determine the canonical utterance.

[0020] Issues arise when developing highly accurate semantic parsers that cover a wider range of tasks and applications. For example, the training process includes adding predefined sets of examples of mappings from natural language utterances into programs. An extensive training includes preparing a large number of examples of mapping data.

[0021] As will be discussed in further detail below, the present disclosure relates to systems and methods for automatically generating instructions based on a received natural language utterance using semantic parsing. The semantic parsing utilizes a pre-trained natural language model based on transformers with a high number of parameters (e.g., millions, billions, etc.) and further uses a constrained decoder to generate a canonical utterance with a strict set of grammatical rules that maps efficiently to program instructions. The present disclosure provides for generation of a prompt as input to the pre-trained natural language model. The prompt includes the natural language utterance and one or more exemplary pairs of conversational sentences. The prompt includes examples that are relevant to a natural language query. The pre-trained natural language model predicts a set of recommended utterances by generating a sequence of tokens and probability values associated with the respective tokens in sequence. A constrained decoder determines a canonical utterance by selecting a sequence of words based on the sequence of tokens and probability values. The constrained decoder determines a canonical utterance based on a set of grammatical rules associated with a canonical utterance. In aspects, the set of grammatical rules characterizes translation between the canonical utterance and instructions. Accordingly, the semantic parsing generates one or more instructions based on the canonical utterance for execution. The system executes the generated instructions to perform a command as a task of an application as given by the user.

[0022] In aspects, the auto-regressive natural language model uses a transformer model, which includes attention mechanisms using an encoder and a decoder on a sequence of tokens or words. In an example, an auto-regressive, transformer-based natural language model, such as a generative pre-trained transformer, may include a very large number (e.g., millions or billions) of parameters. The model may be pre-trained, for example, using large data sets consisting of webpages. The auto-regressive natural language model receives a sequence of words as input and outputs a sequence of text that paraphrases the input. One of skill in the art will appreciate that as the number of pre-trained parameters increases, the model becomes more capable of automatically synthesizing, paraphrasing, and/or translating natural language utterances in a wider range of domains with a higher degree of accuracy.

[0023] FIG. 1 illustrates an overview of an example system **100** for performing semantic parsing for the automatic generation of an instruction. System **100** uses one or more pre-trained natural language models to generate a canonical utterance that can be used to generate an instruction for execution. System **100** includes a client device **102**, a client application server **110**, a semantic

parsing server **120**, a network **130**, a prompt manager **140**, a constrained decoder **144**, and a natural language model **150**. The examples shown in FIG. **1** are specific to a calendar application. One of skill in the art, however, will appreciate that the aspects disclosed herein are not limited to specific applications, but can be employed with other types of applications.

[0024] The client device **102** communicates with client application server **110**, which includes one or more sets of instructions to execute as client applications on the client device **102**. In the provided example, client application server **110** includes an utterance receiver **112** and a calendar viewer **114**. In aspects, the client application server **110** may include additional client applications (e.g., a task viewer and an utterance speaker) not shown in FIG. **1**. The one or more sets of instructions in the client application server **110** may provide an interactive user interface through an interactive browser **104** and utterance input and output devices (e.g., a microphone **106** and a speaker **108**) on the client device **102**. The semantic parsing server **120** includes a natural language (NL) utterance receiver **122**, a canonical utterance receiver **124**, an instruction generator **126**, and an instruction processor **128**. The network **130** provides network connectivity among the client device **102**, the client application server **110**, and the semantic parsing server **120**. The prompt manager **140** includes a task-specific sample generator and example input/output pairs and the NL utterance **146**. The natural language model **150** includes an encoder **152**, a decoder **153**, and trained parameters **156**. The calendar server **180** stores and serves calendar schedule data to the semantic parsing server **120** and the client application server **110** over the network **130**. In aspects, the client device **102** may be a smartphone or a smart speaker that provides a digital assistant service. The digital assistant service may receive a natural language utterance in the form of an interactive voice input and provide a response to the interactive voice input. The response may be provided in various different forms, such as a speech and/or text.

[0025] The client device **102** connects with the client application server **110** via the network **130** to execute applications through the interactive browser **104**. The client application server **110** interacts with the client device **102** and the semantic parsing server **120** via the network **130** to perform information queries and retrievals, calendar scheduling, task management, and/or interactions with various other applications. The natural language model **150** may include a pre-trained, auto-regressive natural language model to translate or paraphrase a natural language utterance to an expression in a paraphrased form. In some aspects, the natural language model is not limited to auto-regressive natural language model. Other types of language models with translating or paraphrasing features may also be used.

[0026] In aspects, the client device **102** provides an environment for interactively developing a semantic parser for particular tasks and applications (e.g., a calendar application, a task application, a scheduler, etc.) through rapid prototyping. The client device **102** interactively specifies a task (e.g., a calendar application) for generating instructions and editing exemplar user-computer utterances (e.g., representative exemplar input/output pairs) that are specific to tasks.

[0027] Client device **102** may be a general computer device operable to receive user provided input, e.g., voice input received via microphone **106**, and/or execute an interactive browser **104** operable to search for information over the network **130**. The interactive browser **104** may render a graphical user interface associated with a web browser, for example. In aspects, the client device **102** may communicate over the network **130** with the client application server **110**. The client device **102** includes microphone **106** for receiving sounds including natural language utterances made by the user. The client device **102** may further includes a speaker **108** for providing an auditory response to the natural language utterance, for example, thought responses may be provided in other forms (e.g., text, video, an action) without departing from the scope of this disclosure.

[0028] The client application server **110** receives input from the client device **102** (e.g., a command for looking up a calendar event) which interactively operate the system **100**. The client application server **110** may comprise applications including the utterance receiver **112** and the calendar viewer

114. The calendar viewer **114** may provide a rendering of one or more calendar schedules for viewing by the user. The calendar viewer **114** may query and receive calendar schedules and appointments by communicating with the semantic parsing server **120**, for example. Additionally, or alternatively, the client application server **110** may include applications that relate to other applications (e.g., a task viewer, an e-commerce application, etc.).

[0029] In aspects, the utterance receiver **112** may receive an utterance from the client device **102** and transmit the received utterance as a natural language utterance to the natural language (NL) utterance receiver **122** of the semantic parsing server **120**. The NL utterance receiver **122** then sends the received natural language utterance to the prompt manager **140**. The canonical utterance receiver **124** may receive a canonical utterance from the constrained decoder **144**. The instruction generator **126** generates one or more instructions based on the canonical utterance. In aspects, the instruction generator **126** includes a compiler that compiles the canonical utterance and outputs one or more instructions. The instruction processor **128** processes or executes the instructions. In aspects, an instruction includes a command and one or more parameters for executing the command. For example, an instruction may access a calendar event in a calendar. The instruction processor **128** may interact with the calendar application **182** in the calendar server **180** to look up the calendar event based on the instruction. The instruction processor **128** provides results from executing the instruction to the calendar viewer **114** in the client application server **110**. The calendar viewer **114** may render the calendar event in the client device **102** by displaying the calendar event and/or announcing the calendar event through the speaker **108** of the client device **102**.

[0030] The prompt manager **140** generates a prompt as input to the natural language model **150** and determines a canonical utterance. In particular, the task-specific sample generator **142** generates the prompt based on the natural language utterance. In aspects, a “prompt” is a set of one or more interactions between a user and a computer (i.e., task-specific input/output pairs). The task-specific sample generator generates the prompt, which includes a set of exemplary conversations between the user and the computer during a task. The set of exemplary conversations include sentences that are relevant to the natural language utterance. Accordingly, the prompt provides the natural language model **150** (pre-trained) a set of examples for influencing predicted utterances to be consistent in context with the exemplary pairs of sentences. In aspects, the task-specific sample generator **142** generates an utterance based on existing instructions. In aspects, generating and providing the prompt as input to the natural language model **150** represent a dynamic “few-shot” learning technique. The natural language model **150** analyzes the prompt with a few examples that precede the natural language utterance before predicting a response to the natural language utterance. Thus, the natural language model **150** effectively “learns” exemplary input/output pairs representing the task before predicting utterances in response to the natural language utterance without performing a training process.

[0031] In aspects, the task-specific sample generator **142** requests the natural language model **150** to determine a level of relevance between respective task-specific input/output pairs and the natural language utterance. The task-specific sample generator **142** selects task-specific input/output pairs that are highly relevant (e.g., selecting top five most relevant pairs) to the natural language utterance for generating the example input/output pairs associate with a task and the NL utterance **146** (prompt). A number of input/output pairs in the prompt may depend on a capacity of input allowed by the natural language model **150**. Some natural language models may restrict the capacity of input based on a number of tokens. In aspects, a cost of re-training and/or fine-tuning the natural language model **150** with large numbers of trained parameters is prohibitively expensive. Re-training may include a preparation of a large amount of training data to influence prediction when there are already a very high number (e.g., millions or billions) of trained parameters. Having a few examples may accomplish fine-tuning the natural language model **150** to be specific to a given task. Fine-tuning a large natural language model may result in developing a

specialized version of the large natural language model for each task.

[0032] In aspects, use of the prompt as input to the natural language model **150** improves accuracy of the predicting tokens of the canonical utterance without extensive re-training. Unlike fine-tuning that limits tasks, use of the prompt enables leveraging the large natural language model for performing many different tasks.

[0033] Additionally or alternatively, the disclosed technology includes fine-tuning of the natural language model **150** using predetermined examples. In aspects, the natural language model **150** can allow for fine-tuning via examples rather than dynamically selecting for prompt creation.” In certain aspects, fine-tuning using known examples may be more accurate than use of the prompt when a number of pre-determined examples used for the fine-tuning is substantially larger than a number of examples received via a prompt.

[0034] The natural language model **150** provides paraphrasing and translations of natural language texts and utterance. In one example, the natural language model **150** may be an auto-regressive language model. The auto-regressive language model receives a sequence of words and/or tokens as input and predicts another sequence of words and/or tokens. In aspects, words and/or tokens may be a part of an utterance or a sentence. Additionally, or alternatively, the natural language model **150** translates a natural language utterance in one language into another natural language utterance in another language. In aspects, the natural language model **150** may include billions of trained parameters.

[0035] The natural language model **150** may be based on a transformer. The encoder **152** receives the natural language utterance in the packet from the task-specific sample generator **142**. The encoder encodes a set of words or tokens in the prompt into multi-dimensional vector representations. The decoder **154** decodes the encoded vectors to predict a sequence of words or tokens and generates predicted utterances. The prediction may be based on a set of probability distributions of each word or token, indicating likelihood of each word as particular words or phrases in utterances.

[0036] In aspects, the pre-trained natural language model includes a very high number (e.g., over millions or billions) of trained parameters **156**, thereby accommodating a wide range of domains in analyzing the natural language utterance and predicting utterances as output. In examples, the natural language model **150** may be pre-trained on a dataset of millions of web pages on the Internet. Such a large-scale, transformer-based language model provides a range of features across a variety of domains. The features include, but are not limited to, paraphrasing (or synthesizing) natural language text based on a given natural language utterance or translating a given natural language utterance into a text or an utterance in another language, for example.

[0037] The constrained decoder **144** requests for and receives a set of predicted tokens in sequence as output from the natural language model **150**. In aspects, a token includes at least a part of a word. Then, the constrained decoder **144** iteratively selects one of the set of predicted tokens filtering out those predicted tokens that do not conform to a set of rules associated with canonical utterances. In some aspects, the rules may include grammatical rules for canonical utterances. The rules characterizes constraints and preferences associated with translation between the canonical utterances and instructions. Accordingly, the allowed sequence of words complies with grammatical rules associated with a canonical utterance. In aspects, the constrained decoder **144** analyzes and compares a word-by-word sequence of each predicted utterance against the allowed sequence of words. After determining a canonical utterance, the constrained decoder **144** provides the determined canonical utterance to the canonical utterance receiver **124** in the semantic parsing server **120**.

[0038] The canonical utterance receiver **124** receives the canonical utterance from the constrained decoder **144**. The instruction generator **126** generates instructions based on the canonical utterance. In aspects, generating the instructions include compiling operations of the canonical utterance into instructions based on a mapping between words in the canonical utterance and commands or

parameters of the instructions.

[0039] The instruction processor **128** executes the generated instructions. For example, the instruction may include finding a calendar event with a particular participant in a calendar database and obtaining a user response to a meeting invite for the calendar event. In aspects, executing the instructions by the instruction processor **128** dispatches commands to the calendar application **182** in the calendar server **180** to locate the calendar event based on the parameters. The calendar application **182** returns a response status of the user for the calendar event to the instruction processor **128**.

[0040] The semantic parsing server **120** may transmit results from execution of the instructions to the client application server **110**, which, in turn, may provide the results to the client device **102**. Additionally, or alternatively, the semantic parsing server **120** may provide the result to the calendar viewer **114**, which causes updating the calendar displayed on the interactive browser **104** of the client device.

[0041] In aspects, constrained decoding limits paraphrased variations in the output without narrowing the wide range of domains supported by the autoregressive, transformer-based natural language model. Developing some of traditional NLP or ASR systems includes training a natural language model to a specific task or domain. The present disclosure, on the other hand, leverages advantages of a pre-trained, auto-regressive, transformer-based natural language model with a constrained decoder. The constrained decoder **144** enables generating a canonical utterance, which complies with strict grammatical rules and the allowed sequence of words.

[0042] In aspects, the present disclosure provides canonical paraphrasing (i.e., paraphrasing a received natural utterance to generate a canonical utterance) based on fine-tuning of the transformer-based natural language model. The fine-tuning of the large-scale, transformer-based natural language model may be costly, however. Training of the natural language model may include preparing data with many-to-many mappings of tokens from natural language utterances to predefined sequences of tokens for canonical utterances. In aspects, a fine-tuning may improve accuracy of the canonical paraphrasing to generate canonical utterance in a new domain. Adding a new functionality includes performing a new fine-tuning of the natural language model using a large number of task-specific training data. Because of the training burden, fine-tuning may not necessarily provide a level of efficiency required to perform rapid prototyping to develop a system in a new domain or applications (e.g., a calendar application, an e-commerce application, etc.).

[0043] Additionally, or alternatively, the dynamic few-shot learning technique improves the accuracy of generating canonical utterances that satisfy a predefined set of grammatical forms. The dynamic few-shot learning technique uses training data that may include one or more of the following: a task description, a few exemplar utterances, and/or a prompt of the task. Unlike the fine-tuning that involves training a natural language model, the few-shot learning technique provides bootstrapping of the training data to the natural language model. Unlike the fine-tuning process, which may be time and resource consuming, particularly with respect to model having a large number of parameters, the dynamic few-shot learning technique is suitable for a rapid prototyping system.

[0044] Additionally or alternatively, the disclosed technology may generate a canonical utterance based on an instruction and paraphrase a canonical utterance into a natural language utterance using a natural language model. Accordingly, the disclosed technology may generate a triplet (a natural language utterance, a canonical utterance, and an instruction). The disclosed technology may use the triplet data for training the natural language model.

[0045] As will be appreciated, the various operations, components, devices, applications, features, etc., described with respect to FIG. **1** are not intended to limit the system **100** to being performed by the particular applications and features described. Accordingly, additional configurations may be employed to practice the methods and systems herein and/or features and applications described may be excluded without departing from the methods and systems disclosed herein.

[0046] FIG. 2 illustrates an example sequence for processing utterance based on semantic parsing in accordance with aspects of the present disclosure. A sequence **200** of data transformation for semantic parsing starts with a natural language utterance **202**. The natural language utterance is a user utterance received by the system. For example, the natural language utterance **202** may be a question: “What time am I getting coffee with the team?”

[0047] Aspects of the present disclosure generate the canonical utterance **204** by

[0048] paraphrasing using a constrained decoder. For example, the canonical utterance **204** based on the exemplar natural language utterance may be the following: “What did I set as my response status for the team meeting?” As described above, the canonical utterance **204** is based on a specific set of grammatical rules that is optimized to efficiently and accurately generate a program **206** based on the canonical utterance. In aspects, a required sentence structure for inquiring an event may include “my response status of find event called something like” following by a parameter name (e.g., subject), followed by pairs of parameter name-value. Accordingly, the canonical utterance **204** satisfies the sentence structure rules for being canonical.

[0049] The program **206** may include expressions as computer-executable instructions. For example, the instructions may indicate: (Yield:output (:responseStatus (singleton (:results (FindEventWrapperWithDefaults:constraint (Constraint[Event]:subject (?~=(String\“team meeting\”)))))). The expression includes one or more commands (e.g., “FindEventWrapperWithDefaults”) and parameters with values (e.g., “subject” and “team meeting”). A result of program execution **208** may represent a result of executing the instructions. For example, the result may be “responded.” In aspects, executing the instructions causes the system to communicate with the calendar application on the calendar server to search for a calendar event with the given conditions. The calendar application then provides a result in response to the query.

[0050] FIG. 3 illustrates an example of processing a natural language utterance to generate and execute instructions using semantic parsing in accordance with aspects of the present disclosure. For example, the sequence **300** describes a scenario where a developer operates the system for rapid prototyping a digital assistant application. The client device **302** (used by the developer) transmits **312** a natural language utterance to the semantic parsing server **304**. In aspects, the transmission is in response to receiving the natural language utterance from the developer as a test utterance. The semantic parsing server **304** transmits the received natural language utterance **314** to the prompt generator **306**.

[0051] In aspects, the prompt generator **306** generates a prompt as input to the natural language model. In aspects, a “prompt” includes the received natural language utterance along with input/output utterance pairs. The pairs include representative sets of examples mapping utterances to a corresponding desired output. The prompt includes the natural language utterance as the last user input. A response to the last input (i.e., the natural language utterance) is for a predicted utterance, which is based on output from the natural language model. It is thus currently empty. The prompt generator **306** may generate **316** the exemplary pairs based on previously processed input/output pairs that are relevant to the current task of the natural language utterance.

[0052] The prompt generator **306** transmits the example input/output pairs of utterance and computer output to the natural language model **308** to determine **318** a degree of relevance between respective pairs (or, in some aspects, the respective exemplary inputs) and the received natural language utterance. The natural language model **308** may be pre-trained. The natural language model **308** may respond to the request by transmitting **320** relevance information. In aspects, the relevance information include the exemplar input/output pair sorted in the sequence of relevance of the respective pairs to the natural language utterance. The prompt generator **306** then generates the prompt by selecting the top five most relevant exemplar input/output pairs and appending the natural language utterance after the pairs. The number of the pairs in the prompt may be any number of pairs and not limited to exemplary five discussed herein. In aspects, a maximum number

of tokens allowed as input to the natural language model **308** may limit the number.

[0053] The prompt generator transmits **324** the prompt to the natural language model **308**. By predicting utterances based on the received prompt, the natural language model **308** effectively performs a “few-shot” learning technique to predict utterances. The technique guides the natural language model to predict utterances based on the example pairs associated with the task without actually training the natural language model with training data. The natural language model **308** provides to the constrained decoder **330** choices of predicted tokens with probability values for the respective tokens as output.

[0054] Additionally or alternatively, the disclosed technology includes fine-tuning of the natural language model based on predetermined examples. In aspects, the fine-tuning of the natural language model may eliminate needs for dynamically generating examples for the prompt. In some other aspects, the fine-tuning may be used in conjunction with use of the prompt to further improve accuracy in the predictions.

[0055] The constrained decoder **330** includes determining an utterance that satisfies a constraint or preference on allowed sequences of words in a canonical utterance. The constrained decoder **330** sends a request **325** for predicted words and/or tokens and receives predicted words/tokens **326** from the natural language model **308**. In aspects, the constrained decoder **330** drops one or more predicted words and/or tokens, which do not satisfy the constraint or preferences. The resulting sequence of predicted words and/or tokens is a canonical utterance.

[0056] The constrained decoder **330** provides the canonical utterance to the semantic parsing server **304**. The semantic parsing server **304** generates instructions based on the canonical utterance by compiling operations. In aspects, the compiling operations includes extracting words of the canonical utterance and analyzing grammatical rules associated with the canonical utterance and a syntax of the instructions, mapping the words to commands and/or parameter values of instructions, and generating the instructions based on the syntax of the instructions.

[0057] The semantic parsing server **304** executes the instructions. In aspects, executing the instructions causes the semantic parsing server **304** to transmit **350** commands and parameters to the application server **310** to perform an operation, such as performing a query. For example, the instructions may cause the semantic parsing server **304** to query the application server **310** (e.g., a calendar server) for calendar events that match given conditions, a name of participants, etc.

[0058] The application server **310** receives the instructions and processes the query. The application server **310** then returns a result **352** to the semantic parsing server **304**. For example, the application server may return a date of the calendar event found. The semantic parsing server **304** provides the result **354** to the client device **302** to inform the developer the result. In aspects, the developer may check the result against expected outcome and perform further proceed with the rapid prototyping of the system.

[0059] FIG. 4 illustrates an example of data used for processing a natural language utterance based on semantic parsing in accordance with aspects of the present disclosure. The data structures **400** include four distinct types of data during various stages of processing the natural language utterance and other data. The natural language utterance **402** is the exemplary utterance “What did I set as my response status for the team meeting?” Based upon the received natural language utterance, the prompt generator generates the prompt **404** as input to the natural language model. For example, the prompt **404** includes two exemplary pairs of input/output (i.e., a user utterance and computer output) pairs that represent a task, followed by the natural language utterance as the last input (as shown in bold text). Each pair includes an utterance by a user and another utterance by the computer. For example, the first pair indicates: “User: what is response statue of the meeting; computer: response status of the meeting.” The second pair indicates: “User: when is the meeting; computer: start time of the meeting.” In example, the two pairs are a part of the top-N relevant, task-specific example input/output pairs. A number of the most relevant pairs for inclusion in the prompt may be predetermined. A maximum number of tokens allowed as input to the natural

language model may limit the number. The goal is to obtain a predicted utterance (i.e., a canonical utterance) as a computer output, in response to the natural language utterance in the prompt. [0060] Constrained decoder **406** decodes each of predicted tokens as outputs of the natural language model to determine a canonical utterance. The constrained decoder **406** uses a data structure to filter out predicted tokens that do not satisfy requirements of specific sequence of words in the canonical utterance. For example, the first word must not be “meeting.” The second word after “my” as the first word must not be “reply.” The example shows that a predicted utterance includes a sequence of words response-status-of-find-event satisfies sets of rules that characterizes constraints and preferences associated with translation between canonical utterances and instructions. Accordingly, the canonical utterance **408** shows the identified canonical utterance: “response status of find event something like “team meeting.”

[0061] Instruction **410** includes a set of instructions generated based on the canonical utterance. For example, the instruction indicates: (Yield:output (:responseStatus (singleton (:results (FindEventWrapperWithDefaults: constraint (Constraint[Event]:subject(?~#(String “team meeting”)))))). The instruction **410** reflects commands and parameters as originally requested by the natural language utterance.

[0062] FIG. 5 is an example of a method **500** for generating a program based on a natural language utterance using semantic parsing to generate a canonical utterance in accordance with aspects of the present disclosure. A general order of the operations for the method **500** is shown in FIG. 5.

Generally, the method **500** begins with start operation **502** and ends with end operation **518**. The method **500** may include more or fewer steps or may arrange the order of the steps differently than those shown in FIG. 5. The method **500** can be executed as a set of computer-executable instructions executed by a computer system and encoded or stored on a computer readable medium. Further, the method **500** can be performed by gates or circuits associated with a processor, an ASIC, an FPGA, a SOC or other hardware device. Hereinafter, the method **600** shall be explained with reference to the systems, components, devices, modules, software, data structures, data characteristic representations, signaling diagrams, methods, etc., described in conjunction with FIGS. 1, 2, 3, 4, 6, 7, and 8A-B.

[0063] Following start operation **502**, the method **500** begins with receive operation **504**, which receives a natural language utterance as an input. A user may input the natural language utterance to command a system (e.g., a digital assistant, a smart speaker, etc.) to perform one or more commands. Identify operation **506** identifies task-specific, exemplar input/output pairs. The input/output pair includes a user utterance and a computer utterance associated with the user utterance. In aspects, the identify operation **506** may generate the input/output pairs based on existing sets of instructions. In aspects, the identify operation **506** may include generating an output utterance of the input/output pair based on a previously generated instruction.

[0064] Determine operation **508** determines relevance information of the exemplary input/output pairs with respect to the natural language utterance. In aspects, system may use the natural language model to generate relevance information based on respective the exemplar input/output pairs and the natural language utterance.

[0065] Generate operation **510** generates a prompt. In aspects, the prompt includes top “N” relevant exemplar input/output pairs, followed by the natural language utterance. The number of the top “N” may be predetermined as a number, the top-3, for example. Additionally, or alternatively, the number may be any number other than three. In aspects, the use of the prompt instead of a single natural language utterance as input represents use of the “few-shot” learning technique to influence prediction without training the natural language model. In aspects, the natural language model may be difficult to train and tune with a small amount of data when the natural language model includes over billions of trained parameters.

[0066] Receive operation **512** receives predicted words and/or tokens as an output from the natural language model. In aspects, the output includes, sets of tokens with probability values associated

with each token. The output from the natural language model is a response to receiving the prompt as an input.

[0067] Determine operation **514** determines a canonical utterance based on the predicted utterance based on the constrained decoding. In aspects, the constrained decoding includes iteratively choosing among predicted tokens based upon constraints and/or preferences associated with translatability of the canonical utterance and instructions. The rules may include grammatical rules associated with the canonical utterance and syntactic rules for translating the canonical utterance into an instruction. The resulting utterance is a canonical utterance.

[0068] Generate operation **516** generates instructions based on the canonical utterance. In aspects, the generate operation **516** includes compiling the canonical utterance and generating the instructions. In aspects, method **500** may end with end operation **518**. The receive operation marked as A **530** corresponds to the receive operation **604** as marked as A **630** in FIG. **6**.

[0069] As should be appreciated, operations **502-518** are described for purposes of illustrating the present methods and systems and are not intended to limit the disclosure to a particular sequence of steps, e.g., steps may be performed in different order, additional steps may be performed, and disclosed steps may be excluded without departing from the present disclosure.

[0070] FIG. **6** is an example of a method **600** for generating a program based on a natural language utterance using semantic parsing to generate a canonical utterance in accordance with aspects of the present disclosure. A general sequence of the operations for the method **600** is shown in FIG. **6**.

Generally, the method **600** begins with start operation **602** and ends with end operation **622**. The method **600** may include more or fewer steps or may arrange the order of the steps differently than those shown in FIG. **6**. The method **600** can be executed as a set of computer-executable instructions executed by a computer system and encoded or stored on a computer readable medium. Further, the method **600** can be performed by gates or circuits associated with a processor, an ASIC, an FPGA, a SOC or other hardware device. Hereinafter, the method **600** shall be explained with reference to the systems, components, devices, modules, software, data structures, data characteristic representations, signaling diagrams, methods, etc., described in conjunction with FIGS. **1, 2, 3, 4, 5, 7, and 8A-B**.

[0071] Following start operation **602**, the method **600** begins with receive operation **604**, which receives a natural language utterance as an input. A user may input the natural language utterance to command a system (e.g., a digital assistant, a smart speaker, etc.) to perform one or more commands.

[0072] Receive operation **604** receives a natural language utterance as input. In aspects, a digital assistant may receive voice input as a natural language utterance. The digital assistant may be an application on a smartphone, a smart speaker, and/or on a digital appliance with a microphone for audio input. The receive operation **604**, as labeled as A **630** corresponds to the receive operation **504** as labeled as A **530** in FIG. **5**

[0073] Generate operation **606** generates a prompt based on the received natural language utterance and a set of correct exemplar input/output pairs of utterances. The input/output pairs are specific to a task associated with the natural language utterance. In aspects the generate operation **606** generates the correct exemplar input/output pairs based on existing sets of instructions that are relevant to the task as indicated by the natural language utterance. Additionally, or alternatively, the generate operation **606** may use the natural language model to determine relevance information of each of the correct exemplar input/output pairs with respect to the natural language utterance. The generate operation **606** select the top-N (e.g., N being any number) most relevant pairs to include inside the prompt. The natural language utterance follows the pairs in the prompt.

[0074] Transmit operation **608** transmits the prompt to the natural language model as an input. The natural language model predicts utterances based on the pairs and the natural language utterance. In aspects, the natural language model may be a pre-trained, autoregressive natural language model based on transformers. The natural language model may include billions of trained parameters to

predict utterances in response to the natural language utterance.

[0075] Receive operation **610** receives from the natural language model predicted utterances. There may be more than utterances as predicted output. Determine operation **612** determines a canonical utterance by performing constrained decoding on each of the predicted utterance. In aspects, the constrained decoding includes extracting and determining whether respective predicted utterances includes words in a specific sequence. The determine operation **612** determines a canonical utterance when a predicted utterance satisfies all the required sequence of words.

[0076] Generate operation **614** generates instructions based on the canonical utterance. IN aspects, the generate operation **614** includes compiling the canonical utterance into instructions with specific syntax. The compiling includes analyzing a sentence structure of the canonical utterance, extracting each word for mapping against syntax information of the instructions to determine values for respective commands and parameters, and generate the instructions. In aspects, the process of generating the instruction is efficient because the grammatical rules for the canonical utterance are associated with syntactic rules for the instruction. The generate operation **614** as labeled as B **632** corresponds to the generate operation **516** as labeled B **532**.

[0077] Execute operation **616** executes the instructions. In aspects, executing the instructions may cause transmitting commands and parameter values to one or more application servers for processing. For example, a calendar application may receive a query based on the command and the parameters to search for a calendar event based on conditions as specified by the parameters values.

[0078] The receive operation **618** receives a result of executing the instructions. For example, the receive operation **618** may receive a result of querying a start time of a calendar event: “responded.”

[0079] Transmit operation **620** transmits the result of executing the instructions to the client device over the network. In aspects, the developer may use the result to further rapid prototype the digital assistant by adjusting tunable parameters of the system, including grammatical rules for constrained decoding and generating the correct exemplar input/output pairs for generating a prompt. In aspects, method **600** may end with end operation **522**.

[0080] As should be appreciated, operations **602-622** are described for purposes of illustrating the present methods and systems and are not intended to limit the disclosure to a particular sequence of steps, e.g., steps may be performed in different order, additional steps may be performed, and disclosed steps may be excluded without departing from the present disclosure.

[0081] FIG. 7 is a block diagram illustrating physical components (e.g., hardware) of a computing device **700** with which aspects of the disclosure may be practiced. The computing device components described below may be suitable for the computing devices described above. In a basic configuration, the computing device **700** may include at least one processing unit **702** and a system memory **704**. Depending on the configuration and type of computing device, the system memory **704** may comprise, but is not limited to, volatile storage (e.g., random access memory), non-volatile storage (e.g., read-only memory), flash memory, or any combination of such memories. The system memory **704** may include an operating system **705** and one or more program tools **706** suitable for performing the various aspects disclosed herein such. The operating system **705**, for example, may be suitable for controlling the operation of the computing device **700**. Furthermore, aspects of the disclosure may be practiced in conjunction with a graphics library, other operating systems, or any other application program and is not limited to any particular application or system. This basic configuration is illustrated in FIG. 7 by those components within a dashed line **708**. The computing device **700** may have additional features or functionality. For example, the computing device **700** may also include additional data storage devices (removable and/or non-removable) such as, for example, magnetic disks, optical disks, or tape. Such additional storage is illustrated in FIG. 7 by a removable storage device **709** and a non-removable storage device **710**.

[0082] As stated above, a number of program tools and data files may be stored in the system

memory **704**. While executing on the at least one processing unit **702**, the program tools **706** (e.g., an application **720**) may perform processes including, but not limited to, the aspects, as described herein. The application **720** includes a natural language utterance receiver **722**, canonical utterance generator **724**, instruction generator **726**, instruction processor **728**, and response provider **730**, as described in more detail with regard to FIG. 1. The canonical utterance generator **724** generates a prompt and using an auto-regressive natural language model and constrained decoding, instruction generator **726**. Other program tools that may be used in accordance with aspects of the present disclosure may include electronic mail and contacts applications, word processing applications, spreadsheet applications, database applications, slide presentation applications, drawing or computer-aided application programs, etc.

[0083] Furthermore, aspects of the disclosure may be practiced in an electrical circuit comprising discrete electronic elements, packaged or integrated electronic chips containing logic gates, a circuit utilizing a microprocessor, or on a single chip containing electronic elements or microprocessors. For example, aspects of the disclosure may be practiced via a system-on-a-chip (SOC) where each or many of the components illustrated in FIG. 7 may be integrated onto a single integrated circuit. Such an SOC device may include one or more processing units, graphics units, communications units, system virtualization units and various application functionality all of which are integrated (or “burned”) onto the chip substrate as a single integrated circuit. When operating via an SOC, the functionality, described herein, with respect to the capability of client to switch protocols may be operated via application-specific logic integrated with other components of the computing device **700** on the single integrated circuit (chip). Aspects of the disclosure may also be practiced using other technologies capable of performing logical operations such as, for example, AND, OR, and NOT, including but not limited to mechanical, optical, fluidic, and quantum technologies. In addition, aspects of the disclosure may be practiced within a general purpose computer or in any other circuits or systems.

[0084] The computing device **700** may also have one or more input device(s) **712**, such as a keyboard, a mouse, a pen, a sound or voice input device, a touch or swipe input device, etc. The output device(s) **714** such as a display, speakers, a printer, etc. may also be included. The aforementioned devices are examples and others may be used. The computing device **700** may include one or more communication connections **716** allowing communications with other computing devices **750**. Examples of suitable communication connections **716** include, but are not limited to, radio frequency (RF) transmitter, receiver, and/or transceiver circuitry; universal serial bus (USB), parallel, and/or serial ports.

[0085] The term computer readable media as used herein may include computer storage media. Computer storage media may include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, or program tools. The system memory **704**, the removable storage device **709**, and the non-removable storage device **710** are all computer storage media examples (e.g., memory storage). Computer storage media may include RAM, ROM, electrically erasable read-only memory (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other article of manufacture which can be used to store information and which can be accessed by the computing device **700**. Any such computer storage media may be part of the computing device **700**. Computer storage media does not include a carrier wave or other propagated or modulated data signal.

[0086] Communication media may be embodied by computer readable instructions, data structures, program tools, or other data in a modulated data signal, such as a carrier wave or other transport mechanism, and includes any information delivery media. The term “modulated data signal” may describe a signal that has one or more characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, communication media may

include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, radio frequency (RF), infrared, and other wireless media.

[0087] FIGS. **8A** and **8B** illustrate a computing device or mobile computing device **800**, for example, a mobile telephone, a smart phone, wearable computer (such as a smart watch), a tablet computer, a laptop computer, and the like, with which aspects of the disclosure may be practiced. In some aspects, the client utilized by a user (e.g., user of client device **102** in FIG. **1**) may be a mobile computing device. With reference to FIG. **8A**, one aspect of a mobile computing device **800** for implementing the aspects is illustrated. In a basic configuration, the mobile computing device **800** is a handheld computer having both input elements and output elements. The mobile computing device **800** typically includes a display **805** and one or more input buttons **810** that allow the user to enter information into the mobile computing device **800**. The display **805** of the mobile computing device **800** may also function as an input device (e.g., a touch screen display). If included as an optional input element, a side input element **815** allows further user input. The side input element **815** may be a rotary switch, a button, or any other type of manual input element. In alternative aspects, mobile computing device **800** may incorporate more or less input elements. For example, the display **805** may not be a touch screen in some aspects. In yet another alternative aspect, the mobile computing device **800** is a portable phone system, such as a cellular phone. The mobile computing device **800** may also include an optional keypad **835**. Optional keypad **835** may be a physical keypad or a “soft” keypad generated on the touch screen display. In various aspects, the output elements include the display **805** for showing a graphical user interface (GUI), a visual indicator **820** (e.g., a light emitting diode), and/or an audio transducer **825** (e.g., a speaker). In some aspects, the mobile computing device **800** incorporates a vibration transducer for providing the user with tactile feedback. In yet another aspect, the mobile computing device **800** incorporates input and/or output ports, such as an audio input (e.g., a microphone jack), an audio output (e.g., a headphone jack), and a video output (e.g., a HDMI port) for sending signals to or receiving signals from an external device.

[0088] FIG. **8B** is a block diagram illustrating the architecture of one aspect of computing device, a server (e.g., a client application server **110**, a semantic parsing server **120**, a natural language model **150**, and a calendar server **180** in FIG. **1**), a mobile computing device, etc. That is, the mobile computing device **800** can incorporate a system **802** (e.g., a system architecture) to implement some aspects. The system **802** can implemented as a “smart phone” capable of running one or more applications (e.g., browser, e-mail, calendaring, contact managers, messaging clients, games, and media clients/players). In some aspects, the system **802** is integrated as a computing device, such as an integrated digital assistant (PDA) and wireless phone.

[0089] One or more application programs **866** may be loaded into the memory **862** and run on or in association with the operating system **864**. Examples of the application programs include phone dialer programs, e-mail programs, information management (PIM) programs, word processing programs, spreadsheet programs, Internet browser programs, messaging programs, and so forth. The system **802** also includes a non-volatile storage area **868** within the memory **862**. The non-volatile storage area **868** may be used to store persistent information that should not be lost if the system **802** is powered down. The application programs **866** may use and store information in the non-volatile storage area **868**, such as e-mail or other messages used by an e-mail application, and the like. A synchronization application (not shown) also resides on the system **802** and is programmed to interact with a corresponding synchronization application resident on a host computer to keep the information stored in the non-volatile storage area **868** synchronized with corresponding information stored at the host computer. As should be appreciated, other applications may be loaded into the memory **862** and run on the mobile computing device **800** described herein.

[0090] The system **802** has a power supply **870**, which may be implemented as one or more batteries. The power supply **870** might further include an external power source, such as an AC adapter or a powered docking cradle that supplements or recharges the batteries.

[0091] The system **802** may also include a radio interface layer **872** that performs the function of transmitting and receiving radio frequency communications. The radio interface layer **872** facilitates wireless connectivity between the system **802** and the “outside world,” via a communications carrier or service provider. Transmissions to and from the radio interface layer **872** are conducted under control of the operating system **864**. In other words, communications received by the radio interface layer **872** may be disseminated to the application programs **866** via the operating system **864**, and vice versa.

[0092] The visual indicator **820** (e.g., LED) may be used to provide visual notifications, and/or an audio interface **874** may be used for producing audible notifications via the audio transducer **825**. In the illustrated configuration, the visual indicator **820** is a light emitting diode (LED) and the audio transducer **825** is a speaker. These devices may be directly coupled to the power supply **870** so that when activated, they remain on for a duration dictated by the notification mechanism even though the processor **860** and other components might shut down for conserving battery power. The LED may be programmed to remain on indefinitely until the user takes action to indicate the powered-on status of the device. The audio interface **874** is used to provide audible signals to and receive audible signals from the user. For example, in addition to being coupled to the audio transducer **825**, the audio interface **874** may also be coupled to a microphone to receive audible input, such as to facilitate a telephone conversation. In accordance with aspects of the present disclosure, the microphone may also serve as an audio sensor to facilitate control of notifications, as will be described below. The system **802** may further include a video interface **876** that enables an operation of an on-board camera **830** to record still images, video stream, and the like.

[0093] A mobile computing device **800** implementing the system **802** may have additional features or functionality. For example, the mobile computing device **800** may also include additional data storage devices (removable and/or non-removable) such as, magnetic disks, optical disks, or tape. Such additional storage is illustrated in FIG. **8B** by the non-volatile storage area **868**.

[0094] Data/information generated or captured by the mobile computing device **800** and stored via the system **802** may be stored locally on the mobile computing device **800**, as described above, or the data may be stored on any number of storage media that may be accessed by the device via the radio interface layer **872** or via a wired connection between the mobile computing device **800** and a separate computing device associated with the mobile computing device **800**, for example, a server computer in a distributed computing network, such as the Internet. As should be appreciated such data/information may be accessed via the mobile computing device **800** via the radio interface layer **872** or via a distributed computing network. Similarly, such data/information may be readily transferred between computing devices for storage and use according to well-known data/information transfer and storage means, including electronic mail and collaborative data/information sharing systems.

[0095] The description and illustration of one or more aspects provided in this application are not intended to limit or restrict the scope of the disclosure as claimed in any way. The claimed disclosure should not be construed as being limited to any aspect, for example, or detail provided in this application. Regardless of whether shown and described in combination or separately, the various features (both structural and methodological) are intended to be selectively included or omitted to produce an embodiment with a particular set of features. Having been provided with the description and illustration of the present application, one skilled in the art may envision variations, modifications, and alternate aspects falling within the spirit of the broader aspects of the general inventive concept embodied in this application that do not depart from the broader scope of the claimed disclosure.

[0096] The present disclosure relates to systems and methods of automatically generating an instruction based on a natural language utterance using a pre-trained natural language model according to at least the examples provided in the sections below. The method comprises receiving the natural language utterance; generating, based at least on a combination of the received natural

language utterance and one or more exemplary pairs of utterances, a canonical utterance using the pre-trained natural language model, wherein the canonical utterance includes, at least in part, a sequence of words based at least on predicted words using the pre-trained natural language model; generating, based on the generated canonical utterance, an instruction; executing the instruction; generating, based on a result of the executing the instruction, a response; and transmitting the response. The pre-trained natural language model includes an auto-regressive natural language model based on a transformer for paraphrasing a first sequence of tokens as a second sequence of tokens, and wherein the first sequence of tokens and the second sequence of tokens are distinct. The canonical utterance is based on one or more of: a set of grammatical rules associated with constraints and preferences for translating between the canonical utterance and the instruction. The method further includes selecting the one or more exemplary pairs of utterances, wherein each exemplary pair of utterances includes an input utterance and an output utterance, and wherein the output utterance is based at least on one of previously generated instructions; dynamically generating, based on the received natural language utterance, a prompt, wherein the prompt includes the one or more exemplary pairs of utterances followed by the received natural language utterance; transmitting the prompt; receiving at least a plurality of a predicted part of a word as output from the pre-trained natural language model; and generating, based on a series of the plurality of a predicted part of a word, the canonical utterance. The pre-trained natural language model includes at least a million trained parameters. Executing the instruction causes transmitting a command to an application server. The canonical utterance is an intermediate expression of automatically translating the natural language utterance into the instruction. The method further includes generating, based on a previously generated instruction, an output utterance; generating, based at least on the output utterance, an input utterance, wherein a combination of the input utterance and the output utterance forms the one or more exemplary pairs of utterances; generating a prompt based on the one or more exemplary pairs of utterances and the natural language utterance; receiving a series of predicted parts of words from the pre-trained natural language model, wherein the predicted parts of the words are associated with the prompt; and generating, based at least on the series of predicted parts of words, the canonical utterance. The method further includes receiving, from the pre-trained natural language model, at least a part of a predicted utterance; iteratively matching a sequence predicted parts of words based on one or more grammatical rules, wherein the one or more grammatical rules correspond to constraints and preferences for translating between the canonical utterance and the instruction; and determining, based on the iteratively matched sequence of predicted parts of words, the canonical utterance.

[0097] Another aspect of the technology relates to a system for automatically generating a program based on a natural language utterance using a pre-trained language model. The system comprises a processor; and a memory storing computer-executable instructions that when executed by the processor cause the system to: receive the natural language utterance; generate, based at least on a combination of the received natural language utterance and one or more exemplary pairs of utterances, a canonical utterance using the pre-trained natural language model, wherein the canonical utterance includes, at least in part, a sequence of words as predicted using the natural language model; generate, based on the generated canonical utterance, an instruction; execute the instruction; generate, based on a result of the executed instruction, a response; and transmit the response. The pre-trained natural language model includes an auto-regressive natural language model based on a transformer for paraphrasing a first sequence of tokens to generate a second sequence of tokens, and wherein the first sequence of tokens and the second sequence of tokens are distinct. The computer-executable instruction when executed further causing the system to: selecting the one or more exemplary pairs of utterances, wherein each pair of exemplar utterances includes an input utterance and an output utterance, and wherein the output utterance is based at least on one of previously generated instructions; dynamically generate, based on the received natural language utterance, a prompt, wherein the prompt includes the one or more exemplary pairs

of utterances followed by the received natural language utterance; transmit the prompt; receive at least a plurality of a predicted part of a word as output from the pre-trained natural language model; and generate, based on a series of the predicted part of a word, the canonical utterance. The computer-executable instruction when executed further causing the system to: generate, based on a previously generated instruction, an output utterance; generate, based at least on the output utterance, an input utterance, wherein a combination of the input utterance and the output utterance forms the one or more exemplary pairs of utterances; generate a prompt based on the one or more exemplary pairs of utterances and the natural language utterance; receive a predicted utterance from the pre-trained natural language model, wherein the predicted utterance is associated with the prompt; and generate, based a series of predicted words from the pre-trained natural language model, the canonical utterance. The computer-executable instruction when executed further causing the system to: receive, from the pre-trained natural language model, at least a part of a predicted utterance; iteratively match a sequence predicted parts of words based on one or more grammatical rules, wherein the one or more grammatical rules correspond to constraints and preferences for translating between the canonical utterance and the instruction; and determine, based on the iteratively matched sequence of predicted parts of words, the canonical utterance.

[0098] In still further aspects, the technology relates to a computer-readable recording medium storing computer-executable instructions for automatically generating a program based on a natural language utterance using a pre-trained language model. The computer-executable instructions, when executed by a processor, cause a computer system to: receive the natural language utterance; generate, based at least on a combination of the received natural language utterance and one or more exemplary pairs of utterances, a canonical utterance using the pre-trained natural language model, wherein the canonical utterance includes, at least in part, a sequence of words as predicted using the natural language model; generate, based on the generated canonical utterance, an instruction; execute the instruction; generate, based on a result of the executed instruction, a response; and transmit the response. The pre-trained natural language model includes an autoregressive natural language model based on a transformer for paraphrasing a first sequence of tokens to generate a second sequence of tokens, and wherein the first sequence and the second sequence are distinct. The canonical utterance is based on one or more of a set of grammatical rules associated with constraints and preferences for translating between the canonical utterance and the instruction. The computer-executable instruction when executed further causing the system to: select the one or more exemplary pairs of utterances, wherein each pair of exemplar utterances includes an input utterance and an output utterance, and wherein the output utterance is based at least on one of previously generated instructions; dynamically generate, based on the received natural language utterance, a prompt, wherein the prompt includes the one or more exemplary pairs of utterances followed by the received natural language utterance; transmit the prompt; receive at least a plurality of a predicted part of a word as output from the pre-trained natural language model; and generate, based on a series of the plurality of a predicted part of words from the pre-trained natural language model, the canonical utterance. The computer-executable instruction when executed further causing the system to: generate, based on a previously generated instruction, an output utterance; generate, based at least on the output utterance, an input utterance, wherein a combination of the input utterance and the output utterance forms the one or more exemplary pairs of utterances; generate a prompt based on the one or more exemplary pairs of utterances and the natural language utterance; receive a series of predicted parts of words from the pre-trained natural language model, wherein the predicted parts of the words are is associated with the prompt; and generate, based at least on the series of predicted parts of words, the canonical utterance.

[0099] The computer-executable instruction when executed further causing the system to: receive, from the pre-trained natural language model, at least a part of a predicted utterance; iteratively match a sequence of predicted parts of words based on one or more grammatical rules, wherein the one or more grammatical rules correspond to constraints and preferences for translating between

the canonical utterance and the instruction; and determine, based on the iteratively matched sequence of predicted parts of words, the canonical utterance.

[0100] Any of the one or more above aspects in combination with any other of the one or more aspect. Any of the one or more aspects as described herein.

Claims

1-20. (canceled)

21. A computer-implemented method of automatically generating an instruction code based on a natural language utterance using a pre-trained natural language model, the method comprising: receiving the natural language utterance; identifying a pair of a query utterance and an answer utterance, wherein the query utterance is distinct from the received natural language utterance; generating a canonical utterance using the pre-trained natural language model, wherein the pre-trained natural language model predicts words based on the received natural language utterance, the identified pair of the query utterance and the answer utterance represents an example of a natural language utterance input and output to influence predicting the words by the pre-trained natural language model without additional training of the pre-trained natural language model, and the canonical utterance includes, at least in part, a sequence of words predicted by the pre-trained natural language model and according to a predetermined constraint of the sequence of words; generating, based on the generated canonical utterance, the instruction code, wherein the instruction code is executable by a processor; and executing, by the processor, the instruction code to generate a response to the instruction code.

22. The method of claim 21, wherein the pre-trained natural language model comprises an autoregressive natural language model based on a transformer for paraphrasing a first sequence of tokens as a second sequence of tokens, and the first sequence of tokens and the second sequence of tokens are distinct.

23. The method of claim 21, wherein the canonical utterance is based on one or more of: a set of grammatical rules associated with constraints and preferences for translating between the canonical utterance and the instruction code.

24. The method of claim 21, further comprising: selecting the pair of the query utterance and the answer utterance, wherein the pair of the query utterance and the answer utterance includes an output utterance from the pre-trained natural language model, and the output utterance is based at least on one of previously generated instructions by the pre-trained natural language model; dynamically generating, based on the received natural language utterance, a prompt, wherein the prompt includes the pair of the query utterance and the answer utterance followed by the received natural language utterance; transmitting the prompt; receiving at least a plurality of a predicted part of a word as output from the pre-trained natural language model; and generating, based at least on the plurality of a predicted part of a word, the canonical utterance.

25. The method of claim 21, wherein the answer utterance represents a previously predicted canonical utterance as predicted by the pre-trained natural language model.

26. The method of claim 21, wherein executing the instruction code comprises an instruction to transmit a command to an application server.

27. The method of claim 21, wherein the canonical utterance is an intermediate expression of automatically translating the received natural language utterance into the instruction code.

28. The method of claim 21, the further comprising: generating, based on a previously generated instruction by the pre-trained natural language model, an output utterance; generating, based at least on the output utterance, an input utterance, wherein a combination of the input utterance and the output utterance forms the pair of the query utterance and the answer utterance; generating a prompt based on the pair of the query utterance and the answer utterance and the received natural language utterance; receiving a series of predicted parts of words from the pre-trained natural

language model, wherein the predicted parts of the words are associated with the prompt; and generating, based at least on the series of predicted parts of words, the canonical utterance.

29. The method of claim 21, further comprising: receiving, from the pre-trained natural language model, at least a part of a predicted utterance; iteratively matching a sequence of predicted parts of words based on one or more grammatical rules, wherein the one or more grammatical rules correspond to constraints and preferences for translating between the canonical utterance and the instruction code; and determining, based on the iteratively matched sequence of predicted parts of words, the canonical utterance.

30. A system for automatically generating an instruction code based on a natural language utterance using a pre-trained natural language model, the system comprises: one or more processors and a memory storing computer-executable instructions that when executed by the one or more processors cause the system to perform operations comprising: receive the natural language utterance; identify a pair of a query utterance and an answer utterance, wherein the query utterance is distinct from the received natural language utterance; generate a canonical utterance using the pre-trained natural language model, wherein the pre-trained natural language model predicts one or more words based on the received natural language utterance, the identified pair of the query utterance and the answer utterance represents an example of a natural language utterance input and output to influence predicting the words by the pre-trained natural language model without further training the pre-trained natural language model, and the canonical utterance includes, at least in part, a sequence of words predicted by the pre-trained natural language model and according to a predetermined constraint of the sequence of words; generate, based on the generated canonical utterance, the instruction code, wherein the instruction code is executable by a processor; and execute, by the processor, the instruction code to generate a response to the instruction code.

31. The system of claim 30, wherein the pre-trained natural language model comprises an auto-regressive natural language model based on a transformer for paraphrasing a first sequence of tokens to generate a second sequence of tokens, and the first sequence of tokens and the second sequence of tokens are distinct.

32. The system of claim 30, wherein the computer-executable instructions, when executed by the one or more processors, further causes the system to perform operations comprising: select the pair of the query utterance and the answer utterance, wherein the pair of the query utterance and the answer utterance includes an output utterance from the pre-trained natural language model, and the output utterance is based at least on one of previously generated instructions by the pre-trained natural language model; dynamically generate, based on the received natural language utterance, a prompt, wherein the prompt includes the pair of the query utterance and the answer utterance followed by the received natural language utterance; transmit the prompt; receive at least a plurality of a predicted part of a word as output from the pre-trained natural language model; and generate, based at least on the plurality of a predicted part of a word, the canonical utterance.

33. The system of claim 30, wherein the computer-executable instructions, when executed by the one or more processors, further causes the system to perform operations comprising: generate, based on a previously generated instruction, an output utterance; generate, based at least on the output utterance, an input utterance, wherein a combination of the input utterance and the output utterance forms the pair of the query utterance and the answer utterance; generate a prompt based on the pair of the query utterance and the answer utterance and the received natural language utterance; receive a predicted utterance from the pre-trained natural language model, wherein the predicted utterance is associated with the prompt; and generate, based a series of predicted words from the pre-trained natural language model, the canonical utterance.

34. The system of claim 30, wherein the computer-executable instructions, when executed by the one or more processors, further causes the system to perform operations comprising: receive, from the pre-trained natural language model, at least a part of a predicted utterance; iteratively match a sequence of predicted parts of words based on one or more grammatical rules, wherein the one or

more grammatical rules correspond to constraints and preferences for translating between the canonical utterance and the instruction code; and determine, based on the iteratively matched sequence of predicted parts of words, the canonical utterance.

35. A computer-readable medium storing computer-executable instructions for automatically generating an instruction code based on a natural language utterance using a pre-trained natural language model, which, when executed by one or more processors, cause a computer system to perform operations comprising: receive the natural language utterance; identify a pair of a query utterance and an answer utterance, wherein the query utterance is distinct from the received natural language utterance; generate a canonical utterance using the pre-trained natural language model, wherein the pre-trained natural language model predict words based on the received natural language utterance, the identified pair of the query utterance and the answer utterance represents an example of a natural language utterance input and output to influence predicting the words by the pre-trained natural language model without further training the pre-trained natural language model, and the canonical utterance includes, at least in part, a sequence of words predicted by the pre-trained natural language model and according to a predetermined constraint of the sequence of words; generate, based on the generated canonical utterance, the instruction code, wherein the instruction code is executable by a processor; and execute, by the processor, the instruction code to generate a response to the instruction code.

36. The computer-readable medium of claim 35, wherein the pre-trained natural language model includes an auto-regressive natural language model based on a transformer for paraphrasing a first sequence of tokens to generate a second sequence of tokens, and wherein the first sequence of tokens and the second sequence of tokens are distinct.

37. The computer-readable medium of claim 35, wherein the canonical utterance is based on one or more of: a set of grammatical rules associated with constraints and preferences for translating between the canonical utterance and the instruction code.

38. The computer-readable medium of claim 35, wherein the computer-executable instructions, when executed by the one or more processors, further causes the system to perform operations comprising: select the pair of the query utterance and the answer utterance, wherein the pair of the query utterance and the answer utterance comprises an output utterance from the pre-trained natural language model, and wherein the output utterance is based at least on one of previously generated instructions; dynamically generate, based on the received natural language utterance, a prompt, wherein the prompt includes the pair of the query utterance and the answer utterance followed by the received natural language utterance; transmit the prompt; receive at least a plurality of a predicted part of a word as output from the pre-trained natural language model; and generate, based on a series of the plurality of a predicted part of words from the pre-trained natural language model, the canonical utterance.

39. The computer-readable medium of claim 35, wherein the computer-executable instructions, when executed by the one or more processors, further causes the system to perform operations comprising: generate, based on a previously generated instruction, an output utterance; generate, based at least on the output utterance, an input utterance, wherein a combination of the input utterance and the output utterance forms the pair of the query utterance and the answer utterance; generate a prompt based on the pair of the query utterance and the answer utterance and the received natural language utterance; receive a series of predicted parts of words from the pre-trained natural language model, wherein the predicted parts of the words are associated with the prompt; and generate, based at least on the series of predicted parts of words, the canonical utterance.

40. The computer-readable medium of claim 35, wherein the computer-executable instructions, when executed by the one or more processors, further causes the system to perform operations comprising: receive, from the pre-trained natural language model, at least a part of a predicted utterance; iteratively match a sequence of predicted parts of words based on one or more

grammatical rules, wherein the one or more grammatical rules correspond to constraints and preferences for translating between the canonical utterance and the instruction code; and determine, based on the iteratively matched sequence of predicted parts of words, the canonical utterance.
